

Signe Aalborg

Senior Production Engineer

(515) 287-4429 | signe.aalborg@emailwire.com | Des Moines, IA

EXPERIENCE

Senior Production Engineer, Iowa Department of Transportation

January 2023 – Present

- Designed and deployed distributed vehicle inspection scheduling API serving 14 regional offices; handles 8,000+ daily requests with 99.97% uptime SLA
- Migrated legacy inspection database from on-premise SQL Server to managed PostgreSQL on AWS; reduced query latency by 63% and eliminated monthly maintenance windows
- Built real-time monitoring and alerting infrastructure using Prometheus and Grafana; on-call rotation covers 24/7 operations for three production services
- Implemented automated failover for permit issuance system across two availability zones; tested quarterly with scheduled chaos engineering exercises
- Mentored four junior engineers on production debugging, capacity planning, and incident response protocols
- Integrated GPT-4 API for parsing unstructured damage reports from inspection forms; reduced data entry errors by 12% with minimal false positives
- Standardized logging across all public-facing APIs to ELK stack; enabled 90% reduction in mean time to resolution for production incidents

Production Engineer, State of Iowa IT Services

July 2021 – December 2022

- Owned reliability and performance for citizen benefits portal processing 2M+ transactions monthly across multiple state agencies
- Designed horizontal scaling strategy for containerized workloads on Kubernetes; increased peak capacity from 500 to 2,400 requests per second
- Implemented circuit breakers and retry logic for all downstream API integrations; reduced cascading failure incidents by 85%
- Wrote runbooks and playbooks for 47 common incident scenarios; average resolution time dropped from 18 minutes to 4 minutes
- Collaborated with policy team to establish SLA targets; communicated technical trade-offs between cost and reliability to non-technical stakeholders
- Maintained PostgreSQL cluster across six nodes with streaming replication; performed monthly backup validation and disaster recovery drills
- Reduced database connection pool exhaustion incidents from 3–4 per month to zero through connection pooling optimization and query analysis

Systems Engineer, Midwest Municipal Cloud Consortium

March 2020 – June 2021

- Deployed and operated 30+ microservices running on Docker and Kubernetes for regional water, utility, and tax collection systems
- Built request routing layer with intelligent load balancing; reduced p99 latencies by 42% during peak municipal billing cycles
- Managed TLS certificate lifecycle automation across all production endpoints; achieved zero certificate expiration incidents

- Diagnosed and resolved critical database deadlocks affecting permit processing system; implemented locking strategy review process
- Documented infrastructure-as-code patterns for 12 cities adopting the platform; conducted training sessions on Terraform and deployment pipelines
- Established metrics collection and retention policy; balanced storage costs against operational visibility needs

Junior Systems Administrator, City of Des Moines Information Services

November 2018 – February 2020

- Provisioned and maintained 25+ Linux servers hosting municipal services and web applications
- Automated daily backup procedures using Bash scripts; reduced manual intervention by 80%
- Supported network and security teams in deploying firewall rules and VPN access for remote city employees
- Responded to security incidents including ransomware detection and containment

TECHNICAL SKILLS

- **Languages & Scripting:** Python, Bash, Go, SQL
- **Cloud & Infrastructure:** AWS (EC2, RDS, S3, Lambda), Docker, Kubernetes, Terraform
- **Databases:** PostgreSQL, MySQL, Redis
- **Observability:** Prometheus, Grafana, ELK Stack, CloudWatch
- **APIs & Integration:** REST, gRPC, event-driven architecture, webhook handling
- **Incident Management:** PagerDuty, on-call rotations, postmortem facilitation

EDUCATION

B.S. in Computer Science

University of Wisconsin–Madison, 2018